

Big Project

→ Domains to work with

↳ FEA

↳ Heat Conduction (+ Elasticity)
?

↳ PMOR

↳ Convective HT (if possible)

↳ Chemical kinetics (if possible)

→ Objectives to achieve

① Simulate heat brick stack as close as possible

② Verify only conduction via exp. using a test setup and Al sheet

③ Build Rom, optimize model, optimize geometry

④ Do DFMEA study of sgs.

→ Software stack for meshing

↳ ① Native MATLAB - PDE
toolbox : not much mesh
control

↳ ② Gmsh + MATLAB

↳ ③ FEA Tool Multiphysics
(GUI based)

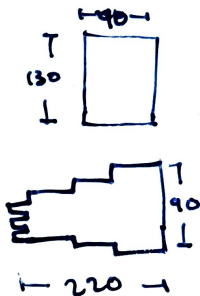
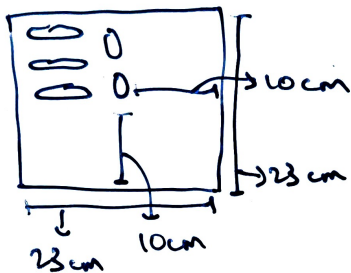
→ Governing equations

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + q_{src}$$

↳ thermal

$$\nabla \cdot (\sigma \nabla u) = 0 - \text{elasticity}$$

> Al 5052 plate dimensions



Big Project

Governing eq.:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \dot{q}$$

$\rho, C_p, k \rightarrow \text{scalar}$

$$T \rightarrow T(x, y)$$

$$\nabla T \rightarrow \begin{bmatrix} \partial T / \partial x \\ \partial T / \partial y \end{bmatrix}$$

$$k \nabla^2 T \rightarrow k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

Integration by parts in 2D

$$\begin{aligned} \int_{\Omega} T_u (\nabla \cdot \nabla T_v) d\Omega &= \int_{\partial\Omega} T_u (\nabla T_v \cdot \hat{n}) d\Gamma \\ &\quad - \int_{\Omega} \nabla T_u \cdot \nabla T_v d\Omega \end{aligned}$$

Variation^deq. for transient process:

$$\int_{\Omega} \rho C_p \frac{\partial T_v}{\partial t} T_v = \int_{\Omega} k \nabla \cdot \nabla T_u T_v + \int_{\Omega} \dot{q} T_v$$

Int. by parts:

$$\rho C_p \int_{\Omega} \frac{\partial T_u}{\partial t} T_v d\Omega = k \left[\int_{\partial\Omega} (\nabla T_u \cdot \hat{n}) T_v d\Gamma - \int_{\Omega} \nabla T_u \cdot \nabla T_v d\Omega \right]$$

$$(\tau_v = 0 \text{ on } \partial\Omega_D) \quad \left(\int_{\Omega} \dot{q} T_v d\Omega \right)$$

$$\therefore \rho C_p \int_{\Omega} \frac{\partial T_u}{\partial t} T_v d\Omega = k \int_{\partial\Omega_N} (\nabla T_u \cdot \hat{n}) T_v d\Gamma + \int_{\Omega} \dot{q} T_v d\Omega - k \int_{\Omega} \nabla T_u \cdot \nabla T_v d\Omega$$

Simplifying and rearranging in
 $a(u, v) = \lambda(v)$ form,

$$\begin{aligned} & \int_{\Omega} \rho \frac{\partial T_u}{\partial t} T_v d\Omega \\ & + k \int_{\Omega} \nabla T_u \cdot \nabla T_v d\Omega = \int_{\Omega} \dot{q} T_v d\Omega \\ & - k \int_{\partial\Omega} (\nabla T_u \cdot \hat{n}) T_v d\Gamma \\ & \quad d\Omega \xrightarrow{\quad} H \text{ (known over } \frac{1}{k} d\Omega) \end{aligned}$$

$$\begin{aligned} & \int_{\Omega} \rho \frac{\partial T_u}{\partial t} T_v d\Omega = \int_{\Omega} \dot{q} T_v d\Omega \\ & + k \int_{\Omega} \nabla T_u \cdot \nabla T_v d\Omega + \int_{\partial\Omega} H T_v d\Gamma \\ & \quad T_v = 0 \text{ on } \partial\Omega_0 \end{aligned}$$

here, $\rho, k, \dot{q}, H \rightarrow$ scalar

$\frac{\partial T_u}{\partial t}, T_v \rightarrow f(x, y)$

$\nabla T_u, \nabla T_v \rightarrow \left[\frac{\partial T_u}{\partial x}, \frac{\partial T_u}{\partial y} \right]_{x,y}$

Here, for a P₁ lagrange triangular element e ,

$$\begin{aligned}\frac{\partial T_u}{\partial t} &\rightarrow \frac{\partial}{\partial t} (a_1(t) N_1^e(x,y) \\ &\quad + a_2(t) N_2^e(x,y) \\ &\quad + a_3(t) N_3^e(x,y)) \\ &= \frac{\partial a_1}{\partial t} N_1^e + \frac{\partial a_2}{\partial t} N_2^e + \frac{\partial a_3}{\partial t} N_3^e\end{aligned}$$

[coefficients of T_u are $f(t)$]

So, can form an eq.,

$$M \ddot{\vec{u}} + K \vec{u} = F$$

where, $\vec{u} = \vec{T}_u \rightarrow [::] \rightarrow$ temp. at nodes
element-wise,

$$M_{ab}^e = \rho C_p \int_{\Omega_e} N_b N_a d\Omega$$

Evaluating integral in M_{ab}^e

$$M_{ab}^e = \int_{\Omega_e} N_b N_a d\Omega$$

($M \rightarrow 3 \times 3$)

Treating N_b as an affine function over Ω_e , $N_b N_a$ is a polynomial of degree 2 in x and y .

Rule:

Integral of a quad. poly. over a triangle = $\frac{1}{3} \times \text{triangle Ar.} \times$
sum of val. of poly. at 3 edge midpt. of $\triangle$

Taking $N_b = f_b$, → value of N_a (1/0 at vertices)

$$M_{ab}^e = \frac{A_e}{3} \left(\frac{f_{b1} + f_{b2}}{2} \frac{\delta_{a1} + \delta_{a2}}{2} + \frac{f_{b2} + f_{b3}}{2} \frac{\delta_{a2} + \delta_{a3}}{2} + \frac{f_{b3} + f_{b1}}{2} \frac{\delta_{a3} + \delta_{a1}}{2} \right) \int_{\Omega_e} d\Omega$$

($f_{b1} \rightarrow f_b$ at vertex 1)
($\delta_{11} = 1$
 $\delta_{12} = 0$)

For $a = 1, 2, 3$

$$M_{:,a}^e = \frac{A_e}{12} \begin{bmatrix} 2f_{a1} + f_{a2} + f_{a3} \\ f_{a1} + 2f_{a2} + f_{a3} \\ f_{a1} + f_{a2} + 2f_{a3} \end{bmatrix}$$

scf

For $b = 1, 2, 3$,

here, $f_{11} = 1$, $f_{12} = 0$ (like δ)

so,

$$M^e = \frac{A_e}{12} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \text{ scf}$$

Assembling M^e : same as that of K^e

Time stepping scheme

$$M \dot{u} = F - K u \quad (u \rightarrow \text{vector of temp at nodes})$$
$$\dot{u} = M^{-1} F - M^{-1} K u$$

Discretizing,

$$\frac{u_{t+h} - u_t}{\Delta t} = M^{-1}F - M^{-1}K \bar{u}_{t+h}$$

$$u_{t+h} - u_t = M^{-1}F \Delta t - M^{-1}K \Delta t u_{t+h}$$

$$(I + M^{-1}K \Delta t) u_{t+h} = u_t + M^{-1}F \Delta t$$

$$u_{t+h} = (I + M^{-1}K \Delta t)^{-1} (u_t + M^{-1}F \Delta t)$$

OR

$$M u_{t+h} - M u_t = F \Delta t - K \Delta t u_{t+h}$$

$$(M + K \Delta t) u_{t+h} = F \Delta t + M u_t$$

$$u_{t+h} = (M + K \Delta t)^{-1} (F \Delta t + M u_t)$$

(avoids M^{-1})

Including convective B.C.

(Robin B.C.)

- ↳ Applied to nodes on plate which will be immersed in water bath
- ↳ For each element subject to convection, avg. T of element nodes T_{avg} is used to calculate vol. heat gen. $\dot{q} = -h(T_{avg} - T_{\infty})/t$ in each element
- ↳ F vector is updated after every timestep to account of changing $\dot{q}$
- ↳ F vector updation is inefficient, try finding other optimal technique